

CellChatAccelRcpp (sparse_stream) accelerates CellChat by an order of magnitude with zero loss of accuracy

Speed & memory (A–C, G): median 17.5× faster than original CellChat and 1.9× lower peak memory at scale (216 paired runs).
Biology (D–F, H): a representative glioma sample (9 cell groups) reproduces the network and ligand–receptor signalling exactly; equivalence (max $|\Delta|$ = 0, r = 1.0000) holds across all 216 experiments.

